

Raj Tejpal Khatik

AI/ML Engineer — Agentic AI & GenAI Systems

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MSc Applied AI candidate at the University of Warwick, graduating September 2026, with two years eight months of prior industry experience in data analytics and ML engineering. Designs multi-agent, retrieval-augmented systems — from causal-chain equity analysis to hallucination detection — and publishes the research behind each one.

Experience

AI Engineer — H&B Infotech, India

March 2024 – March 2025

Built ML pipelines to prepare and process data for model training, integrated into existing backend systems — improving analysis accuracy and reporting efficiency. Stack: Python, NumPy, Pandas, ML Pipelines.

Software Engineer — H&B Infotech, India

March 2023 – March 2024

Engineered backend systems and data pipelines powering internal reporting tools, with data validation and front-end integration work. Stack: Python, MySQL, Pandas, NumPy.

Software Engineer Intern — H&B Infotech, India

August 2022 – February 2023

Developed and validated Python-MySQL modules to support internal systems. Built front-end interfaces for internal tools. Stack: Python, MySQL, HTML/CSS, JavaScript, React.

Education

MSc Artificial Intelligence — University of Warwick, WMG

September 2025 – September 2026 (in progress)

BEng, Computer Science & Engineering — Gujarat Technological University

August 2021 – May 2025 · CGPA 8.49 / 10

Projects

Six deployed systems, each paired with a published paper, built on a shared architecture — LangGraph orchestration, FAISS retrieval, Streamlit interfaces — applied to a different domain each time.

- **SemiBot (flagship)** — 18-factor causal-chain RAG system for semiconductor equities. Directional accuracy scales from 84.0% (>\$1T market cap) to 41.7% (mid-cap) across 426 observations / 48 tickers (logistic regression, $p < 0.001$). `semibot-raj.streamlit.app`
- **GoldBot** — Nine-factor safe-haven analyzer for gold, stress-tested across four historical regimes. Diagnostic finding: output stayed within a 37–50% band ($SD=4.2$) across 18 events, exposing trend-matching rather than true event discrimination. `goldbot-raj.streamlit.app`
- **SilverBot** — Decomposes silver price behavior into independent industrial and monetary strength scores. Industrial signals beat Monetary in 3 of 4 regimes (61.7% vs 40.0%, $p=0.041$), reversing during the 2020 liquidity crisis. `silverbot-raj.streamlit.app`
- **Hallucination Detector (flagship)** — Six-signal ensemble for LLM hallucination detection (NLI, retrieval grounding, three-model LLM-judge panel, self-consistency). 72.4–92.2% accuracy across three benchmarks. `hallucinationdetecto.streamlit.app`
- **HITL Email Agent** — Human-in-the-loop email drafting agent with LangGraph interrupt/resume, dual Gmail OAuth, and a Manifest V3 Chrome extension. Zero secrets exposed across full repo history.
- **Retail Research Assistant** — Discovery, research, comparison, and portfolio analysis across all 503 S&P 500 constituents. Passed a 100-question adversarial safety test, 100/100. `retailresearch-raj.streamlit.app`

Papers

Four preprints live on SSRN, in active submission to peer-reviewed journals — covering the SemiBot, GoldBot, SilverBot, and Hallucination Detector systems above (SSRN 7199998, 7200198, 7203201, 7200222).

Dissertation

Understanding the Role of Artificial Intelligence in Life Cycle Assessment — supervised by Dr. You Wu, WMG, University of Warwick. Applied unsupervised learning (KMeans, Isolation Forest, PCA) to the ecoinvent 3.11 dataset and compared GPT-4o-mini vs. Claude Haiku 4.5 on structured LCA tasks (Claude: 100% exact-match vs. GPT-4o-mini: 87.0%, n=100, McNemar's $p=0.0002$). Due September 2026.

Skills

LangGraph · LangChain · RAG · FAISS · Agentic AI · Python · Streamlit · FastAPI